DSA03-Natural Language Processing

List of Lab Program

11. Implement a simple top-down parser for context-free grammars using python.

Code:

```
import nltk
from nltk import CFG
from nltk.parse import RecursiveDescentParser
grammar = CFG.fromstring("""
S -> NP VP
NP -> 'the' 'boy' | 'the' 'girl'
VP -> 'plays' 'cricket' | 'plays' 'football'
""")
parser = RecursiveDescentParser(grammar)
sentence = ['the', 'boy', 'plays', 'cricket']
print("Input Sentence:", " ".join(sentence))
print("Parse Tree:")
for tree in parser.parse(sentence):
    print(tree)
    tree.pretty_print()
```

Output:

```
Input Sentence: the boy plays cricket
Parse Tree:
(S (NP the boy) (VP plays cricket))
```

